

Chapter 8: Conservation of Energy

Non-Isolated system

Isolated system

Kinetic friction

ΔE_{mech} for non-conservative forces

Power

1

Introduction

- ▶ Conservation law of a physical quantity
 - Quantity does not change its value with time
- ▶ Law of Conservation of energy
 - The total energy of the universe is unchanged by a physical process.
 - Initial total energy = final total energy
- ▶ We cannot create or destroy energy, we can only change/convert it from one form to another

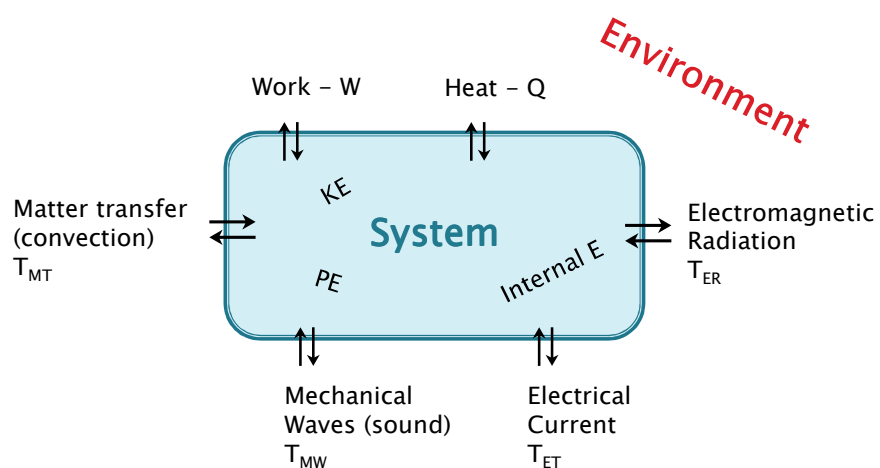
2

Forms of energy

- ▶ A large number of energy forms:
 - Chemical energy
 - Nuclear-energy
 - Elastic energy, to name a few
- ▶ On a fundamental level
 - Kinetic energy
 - Potential energy
 - Internal energy

3

Non-isolated system



4

Non-isolated system

- ▶ If the total amount of energy in a system changes, it can ONLY be because energy has crossed the boundary of the system by one or more transfer mechanisms mentioned previously.

- ▶ $\Delta E_{system} = \sum T$

- ΔE - Total change in the system's energy
- $\sum T$ - sum total of energy transferred over the system boundary via one or more transfer mechanism.
- $T_{work} = W$ - amount of work done.
- $T_{heat} = Q$ - Amount of heat transferred (ch 20)

- ▶
$$\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int}$$

$$= W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

5

Non-isolated system

- ▶ Suppose a force is applied to a non-isolated system and the force moves through a displacement. If the only effect on the system is to change its speed.

- ▶
$$\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int}$$

$$\Delta K + \Delta U + \Delta E_{int} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

- The only transfer mechanism is work.
- Only energy to change will be kinetic energy.
- The above equation then reduces to:

$$\Delta K = W$$

- Work-Kinetic Energy Theorem

6

Isolated systems

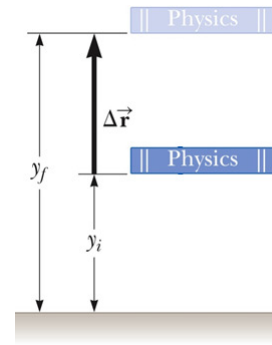
- ▶ Here we choose our system so that no energy crosses the system boundary by any method.

- ▶ Consider book–Earth system.

- After the book has been lifted by an external force– Gravitational potential energy
- $W = \Delta U$ (Can be confirmed with previous ΔE equation.)
- Now consider the work done on the book alone by gravitational force, as book falls to its original height.

$$W_{on\ book} = (m\vec{g}) \cdot \Delta\vec{r} = (-mg\hat{j}) \cdot [(y_f - y_i)\hat{j}]$$

$$= mgy_i - mgy_f$$



7

Isolated systems

- ▶ From Work–kinetic energy theorem:

$$W_{on\ book} = \Delta K_{book}$$

- ▶ Combining these two expressions for the work done on the book:

$$\Delta K_{book} = mgy_i - mgy_f$$

- ▶ Now for the *isolated system* of the book this becomes:

$$\Delta K = -\Delta U_g$$

$$\Delta K + \Delta U_g = 0$$

- Sum of the changes of the stored energy in the system (LHS) = 0, since not energy was transferred across the boundary of the system

- ▶ This is valid for a system with any type of PE

8

Isolated systems

- ▶ This is can be derived a system with any type of PE

$$\Delta K + \Delta U = 0 \quad E_{\text{mech}} = K + U$$

- ▶ We can re-write this as: $\Delta E_{\text{mech}} = 0$

- This is a statement of the conservation of mechanical energy for an isolated system with no non-conservative forces acting.

- ▶ We can write the changes of the energies explicitly:

$$\Delta K + \Delta U = 0$$

$$(K_f - K_i) + (U_f - U_i) = 0$$

$$K_f + U_f = K_i + U_i$$

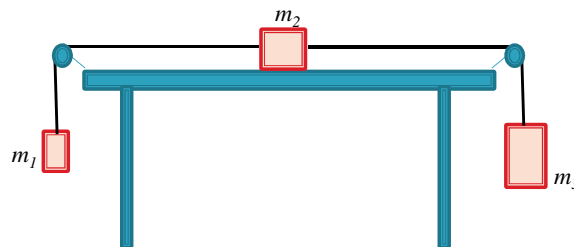
$$\frac{1}{2}mv_f^2 + mgy_f = \frac{1}{2}mv_i^2 + mgy_i$$

$$E_{\text{mech},f} = E_{\text{mech},i}$$

9

Example

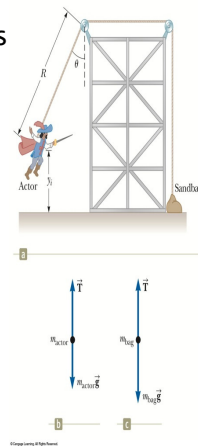
Three objects with masses $m_1 = 5 \text{ kg}$, $m_2 = 10 \text{ kg}$, and $m_3 = 15 \text{ kg}$ are connect with massless strings that pass over two frictionless pulleys as indicated in the figure. The horizontal surface is frictionless, and the system is released from rest. Find the speed of m_3 after it has moved a distance of 0.4m downwards.



10

Example 8.2

A structure that an actor will use to swing across the stage is shown. The actor, 65.0 kg, has a harness which is fastened to a sandbag, 130 kg, by a light-weight cable that passes over two pulleys. The distance between the actor and the first pulley is 3 m. For the system to work successfully, the sandbag must not lift off the floor. What is the maximum initial angle so that the sandbag does not lift off the floor?



Work through problem in textbook

System with kinetic friction

- ▶ Consider an object with mass, m , that moves over a surface with friction.
- ▶ The object moves across a distance d .
- ▶ The work done by the kinetic frictional force and the other forces gives the total work done.

$$\begin{aligned}
 W_{\text{total}} &= \sum W_{\text{other forces}} + \int \vec{f}_k \cdot d\vec{r} \\
 &= \int (\sum \vec{F}_{\text{other forces}}) \cdot d\vec{r} + \int \vec{f}_k \cdot d\vec{r} \\
 &= \int (\sum \vec{F}_{\text{other forces}} + \vec{f}_k) \cdot d\vec{r} \quad \left(\sum \vec{F}_{\text{other forces}} + \vec{f}_k = \sum \vec{F} = \vec{F}_{\text{net}} \right) \\
 W_{\text{total}} &= \int (\sum \vec{F}) \cdot d\vec{r}
 \end{aligned}$$

System with kinetic friction

► But $\sum \vec{F} = \vec{F}_{net} = m\vec{a}$

► And therefore

$$\begin{aligned}
 W_{total} &= \int (m\vec{a}) \cdot d\vec{r} \\
 &= \int \left(m \frac{d\vec{v}}{dt} \right) \cdot d\vec{r} \\
 &= \int \left(m \frac{d\vec{v}}{dt} \right) \cdot \vec{v} dt \quad (d\vec{r} = \vec{v} dt)
 \end{aligned}$$

► Product-rule for derivatives: $\frac{d}{dt}(\vec{v} \cdot \vec{v}) = \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt} = 2 \frac{d\vec{v}}{dt} \cdot \vec{v}$

► It follows that

$$\begin{aligned}
 \frac{d}{dt}(v^2) &= 2 \frac{d\vec{v}}{dt} \cdot \vec{v} \\
 \frac{1}{2} \frac{d}{dt}(v^2) &= \frac{d\vec{v}}{dt} \cdot \vec{v} \quad \textcircled{1}
 \end{aligned}$$

13

System with kinetic friction

► And then

$$\begin{aligned}
 W_{total} &= \int m \frac{d\vec{v}}{dt} \cdot \vec{v} dt \\
 &= \frac{1}{2} m \int_{t_i}^{t_f} \frac{d(v^2)}{dt} dt \quad \text{from } \textcircled{1} \\
 &= \frac{1}{2} m \int_{v_i}^{v_f} 1 d(v^2) \\
 &= \frac{1}{2} m v^2 \Big|_{v_i}^{v_f} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = \Delta K
 \end{aligned}$$

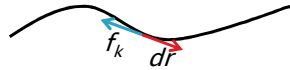
$$W_{total} = \sum W_{other \text{ forces}} + \int \vec{f}_k \cdot d\vec{r} = \Delta K$$

14

System with kinetic friction

$$W_{total} = \sum W_{other\ forces} + \int \vec{f}_k \cdot d\vec{r} = \Delta K$$

- ▶ $\vec{f}_k$ and $d\vec{r}$ are always in opposite directions



- ▶ $\vec{f}_k \cdot d\vec{r} = -f_k dr$
 $\sum W_{other\ forces} + \int \vec{f}_k \cdot d\vec{r} = \Delta K$
 $\sum W_{other\ forces} - \int f_k dr = \Delta K$
 $\sum W_{other\ forces} - f_k \int_0^d dr = \Delta K$

$$\sum W_{other\ forces} - f_k d = \Delta K$$

Conservation of energy law if the system has kinetic frictional force acting on the object

15

System with kinetic friction

- ▶ $\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int} = 0$
- ▶ With no PE present
- ▶ $\Delta E_{system} = \Delta K + 0 + \Delta E_{int} = 0$
- ▶ When the other forces do not do any work:

$$\begin{aligned} \sum W_{other\ forces} &= 0 \\ \Rightarrow -f_k d &= \Delta K \\ \Delta E_{system} &= -f_k d + \Delta E_{int} = 0 \\ \therefore \Delta E_{int} &= f_k d \end{aligned}$$

- ▶ Friction changes the internal energy of the system
- ▶ In general, when there are other forces that do work:

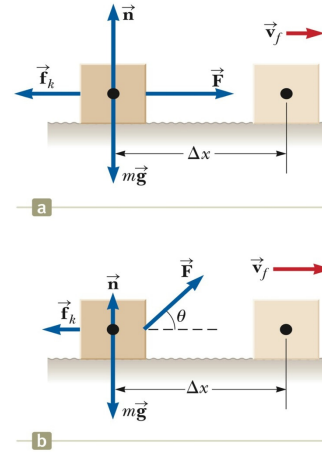
$$\sum W_{other\ forces} = \Delta K + \Delta E_{int}$$

16

Example 8.4

A 6.0 kg block initially at rest is pulled to the right along a horizontal surface by a constant horizontal force of 12 N.

- Find the speed of the block after it has moved 3.0 m if the surfaces in contact have a coefficient of kinetic friction of 0.15.
- Suppose that the force is applied at an angle θ (Figure b). At what angle should the force be applied to achieve the largest possible speed after the block has moved 3.0 m to the right?

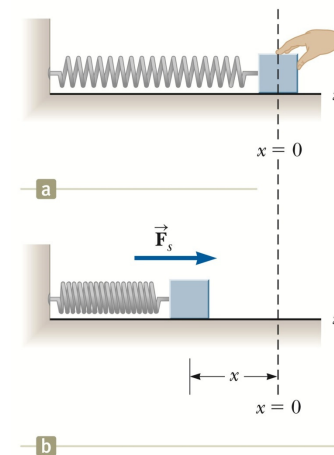


17

Example 8.6

A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 1000 N/m as shown in the figure (a). The spring is compressed 2.0 cm and is then released from rest, figure (b).

- Calculate the speed of the block as it passes through the equilibrium position, $x = 0$ m, if the surface is frictionless.
- Calculate the speed of the block as it passes through the equilibrium position, if a constant friction force of 4.0 N retards the motion during the process.

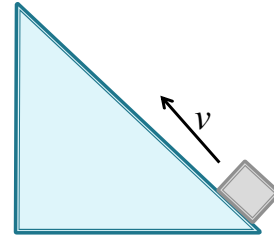


SYSTEM? System is the block

18

Changes in Mechanical Energy for Non-conservative forces

- ▶ We now move a block (object) up a slope.
- ▶ The slope is a rough surface.
- ▶ We also have a change in the gravitational potential energy of the object.



- ▶ Consider the mechanical energy:

$$\Delta E_{mech} = \Delta K + \Delta U_{grav} = -f_k d = -\Delta E_{int}$$

- ▶ And therefore:

$$\Delta E_{system} = \Delta K + \Delta U_{grav} + \Delta E_{int} = 0$$

$$\Rightarrow \Delta K + \Delta U_{grav} = -\Delta E_{int}$$

19

Changes in Mechanical Energy for Non-conservative forces

- ▶ We can generalise the above equation for **any PE**.

$$\Delta K + \Delta U + \Delta E_{int} = 0$$

$$\Delta K + \Delta U = -\Delta E_{int}$$

$$\Delta K + \Delta U = -f_k d$$

- ▶ For a system where non-conservative forces are present:

$$\sum W_{other\ forces} - f_k d = \Delta E_{mech}$$

$$\sum W_{other\ forces} - \Delta E_{int} = \Delta K + \Delta U$$

$$\Rightarrow W = \sum W_{other\ forces} = \Delta K + \Delta U + \Delta E_{int}$$

Where $W \neq 0$, which then means that this is a non-conservative system.

20

Power

- ▶ There are **different ways to transfer energy** to (or from) a system – for example by doing work.
- ▶ If an external force is applied to an object, then work is done during a time interval Δt
- ▶ **The rate at which energy is transferred** is then defined as:

$$P_{avg} = \frac{W}{\Delta t}$$

- ▶ **This is the average power**
- ▶ **The instantaneous power** is then:

$$\lim_{\Delta t \rightarrow 0} \frac{W}{\Delta t} = \frac{dW}{dt} = P$$

21

Power

- ▶ Using this definition and

$$dW = \vec{F} \cdot d\vec{r}$$

$$\Rightarrow P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

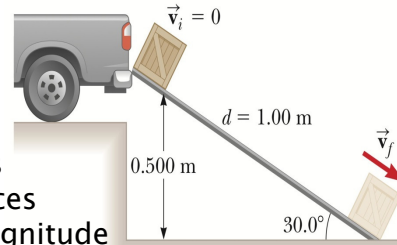
- ▶ The unit for power is: watt

$$1\text{W} = \frac{1\text{J}}{1\text{s}} = 1\text{kg} \cdot \text{m}^2/\text{s}^3$$

22

Example 8.7

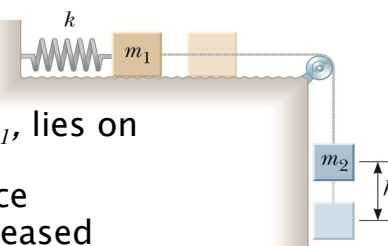
- ▶ A 3.00 kg crate slides down a ramp. The ramp is 1.00 m in length and inclined at an angle of 30.0° . The crate starts from rest at the top, experiences a constant friction force of magnitude 5.00 N, and continues to move a short distance on the horizontal floor after it leaves the ramp.
- ▶ Use energy methods to determine the speed of the crate at the bottom of the ramp.
- ▶ How far does the crate slide on the horizontal floor if it continues to experience a friction force of magnitude 5.00 N?



23

Example 8.9

Two blocks are connected by a light string that passes over a frictionless pulley. The block of mass, m_1 , lies on a horizontal surface and is connected to a spring of force constant k . The system is released from rest when the spring is relaxed. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface.



24

Example

A bungee jumper of mass 61 kg jumps of the edge of the bridge. He is attached to an elastic cord which is originally 25 m long. The cord stretches a distance d when the person is hanging below. The spring constant of the cord is 160 N/m.

- (a) Determine the distance which the cord stretches.
- (b) Determine the net force on the jumper at the lowest point.